

DIGITAL HOSPITAL PATIENT MANAGEMENT SYSTEM

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1.Patient Registration

When a person visits the hospital for the first time
Receptionist opens a new Patient Registration and
They enter:

Basic Information:

- Patient ID
- Full name
- Date of birth
- Age
- Gender
- Phone number
- Email
- Address
- Emergency contact

Medical Information:

- Blood group
- Height
- Weight
- Allergies
- Existing medical conditions
- Previous surgeries
- Current medications

- Emergency medical information

Optional identification

- ABHA ID
- Government ID/reference
- Insurance information

The system then generates a unique:

Patient ID

For example:

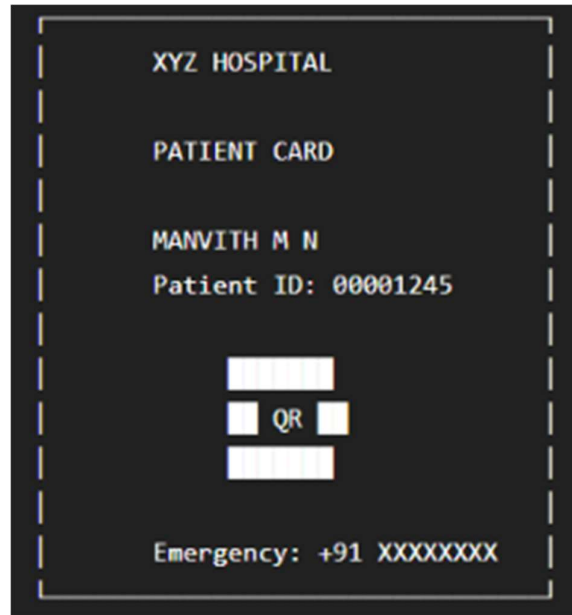
MTB-HOSP-00001245

And generates a **QR code** linked to that patient.

2.QR Patient Card

This is one of the most important parts of the Digital Hospital Patient Management System.

The patient receives:



Physical card

- The QR should contain something like:
<https://hospital.com/patient/secure-token>
- Preferably a non-guessable patient reference/token.
- The server then verifies the request and retrieves the actual patient record.
- That way, if somebody photographs the card, they don't automatically get the person's medical history.

3.Reception QR Scanning

- Patient arrives at hospital.
- Receptionist: **Scan QR** → **Patient identified**
Immediately the reception dashboard shows:

PATIENT PROFILE
Patient ID: MTB-00001245
Name: Manvith M N
Age: 20
DOB: 04/09/2005
Gender: Male
Blood Group: B+
Phone: XXXXXXXX
Address: XXXXX
Allergies: None
Previous Visits: 5
Last Visit: 02 August 2026
Last Doctor: Dr. XYZ
Last Prescription:
View

The receptionist doesn't have to manually search the patient's name/phone number.

4.Doctor Dashboard

- This should be a separate interface.
- Doctor logs in → Doctor Dashboard

Today's Appointments

09:00	Patient A
09:30	Patient B
10:00	Patient C
10:30	Patient D

- When the doctor opens Patient B:

PATIENT PROFILE

Name: Manvith M N	
Age: 20	
Blood Group: B+	
Medical History	
Previous Visits	

02 Aug 2026

Dr. XYZ

Reason: Fever

15 July 2026

Dr. ABC

Reason: Headache

10 June 2026

Dr. XYZ

Reason: General Checkup

- Doctor can then see the patient's historical records.

5.Digital Consultation

- This is another major module.
- During consultation, doctor can enter:

Symptoms

Chief Complaint:

Symptoms:

- ☒ Fever
- ☒ Headache
- ☐ Cough
- ☐ Vomiting

Vitals

Blood Pressure: 120/80

Heart Rate: 78

Temperature: 98.6°F

Weight: 70 kg

SpO₂: 98%

Diagnosis:

Doctor's Notes:

6.Digital Prescription

- Doctor clicks → **Create Prescription**
- Prescription:

```
Prescription

Patient:
Manvith M N

Doctor:
Dr. XYZ

Diagnosis:
Viral Fever

Medicines:

1. Paracetamol
   500 mg
   1-0-1
   3 Days

2. XYZ Tablet
   1-0-0
   5 Days

Instructions:
Take medicines after food.
Drink sufficient water.
```

- Doctor clicks → **Save Prescription**
- The prescription becomes part of the patient's medical history.
- You could also generate a **PDF prescription**.

7.Patient Portal

- This is the other half of the system.
- Patient can log in using:

Option 1

Mobile number + OTP

Option 2

Patient ID + OTP

Then:

Welcome, Manvith

Upcoming Appointment

Dr. XYZ

Cardiology

12 August 2026

10:30 AM

My Health:

Personal Information

Name

DOB

Age

Blood Group

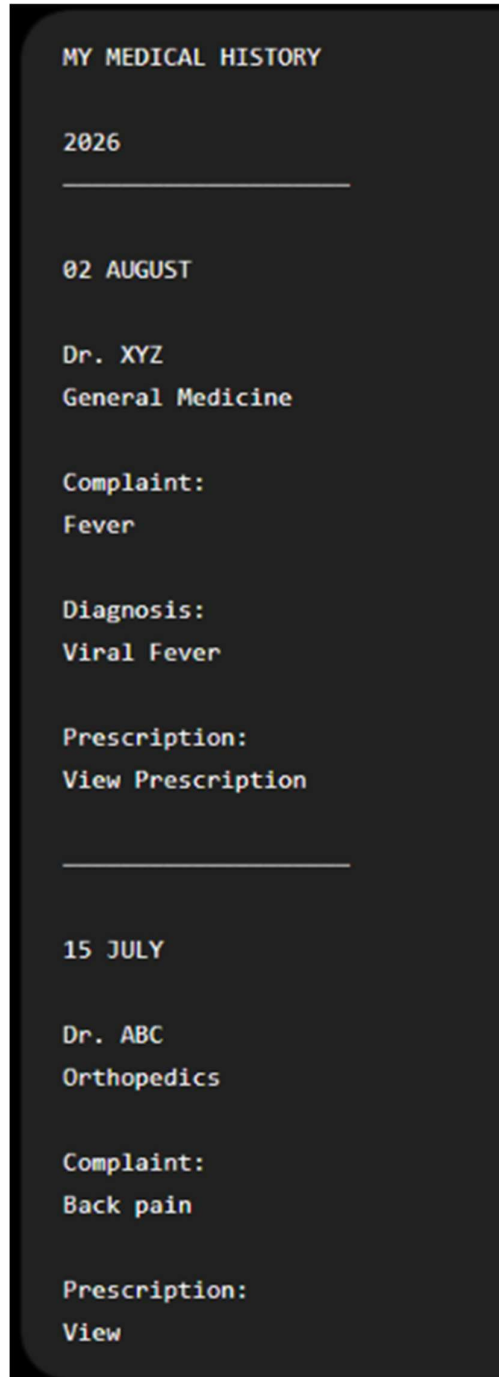
Weight

Height

Allergies

8. Medical History

- Patient should be able to see:



- This is basically the patient's **personal health timeline**.

9. Appointment Booking

- Patient opens → **Book Appointment**

Select:

Department

- General Medicine
- Cardiology
- Dermatology
- Orthopedics
- ENT
- Pediatrics
- Gynecology
- Dentistry

Then:

Doctor

- Dr. XYZ
General Medicine
- Dr. ABC
General Medicine

Then:

Date

- August 12
- August 13
- August 14

Then available slots:

10:00 AM	Available
10:30 AM	Available
11:00 AM	Booked
11:30 AM	Available

Patient selects:

10:30 AM

And books.

10. Appointment Notifications

After booking:



Send notification through:

- SMS
- WhatsApp
- Email
- Push notification

We can later add:

"Your appointment is tomorrow at 10:30 AM."

11.Hospital Admin Dashboard

This becomes the central control panel.

- Admin can manage:

1. Patients

Total Patients

New Patients

Active Patients

Today's Registrations

2. Doctors

Add Doctor

Remove Doctor

Doctor Schedule

Specialization

Availability

3. Appointments

Today's Appointments

Completed

Pending

Cancelled

No-show

4. Departments

General Medicine

Cardiology

Orthopedics

etc.

12.Different User Roles

- This is extremely important → Don't give everybody access to everything.

1. Super Admin

Full access.

Hospital settings

Users

Doctors

Patients

Reports

Billing

System configuration

2. Receptionist

Register patient

Scan QR

View basic patient information

Book appointments

Manage queue

3. Doctor

View patient medical history

Create consultation

Create prescription

View previous prescriptions

Update clinical notes

4. Nurse

Vitals

BP

Weight

Temperature

SpO2

etc.

5. Pharmacist

View prescriptions

Medicine dispensing

6. Patient

Own profile

Own medical records

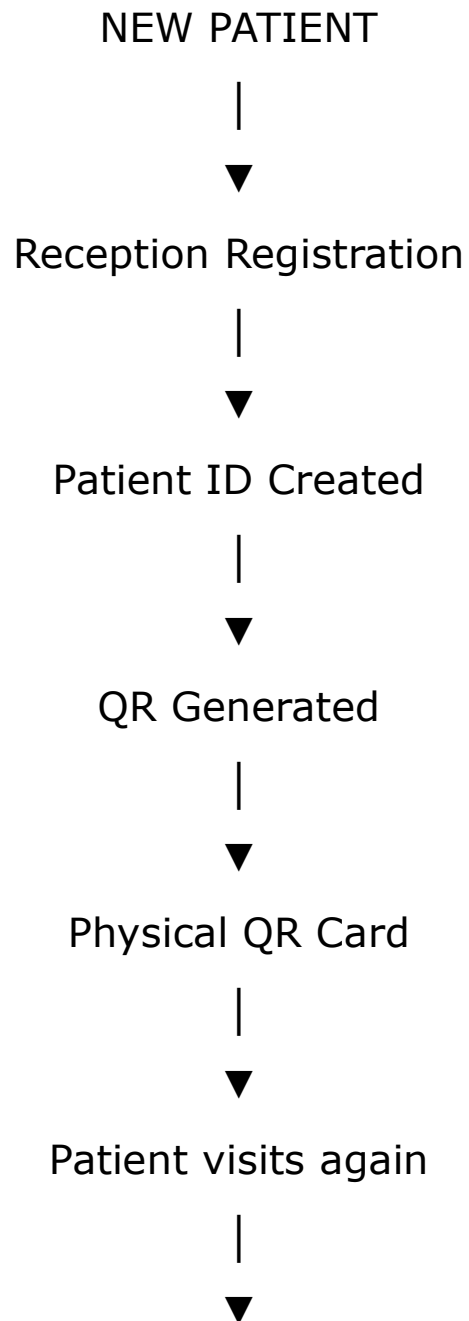
Own prescriptions

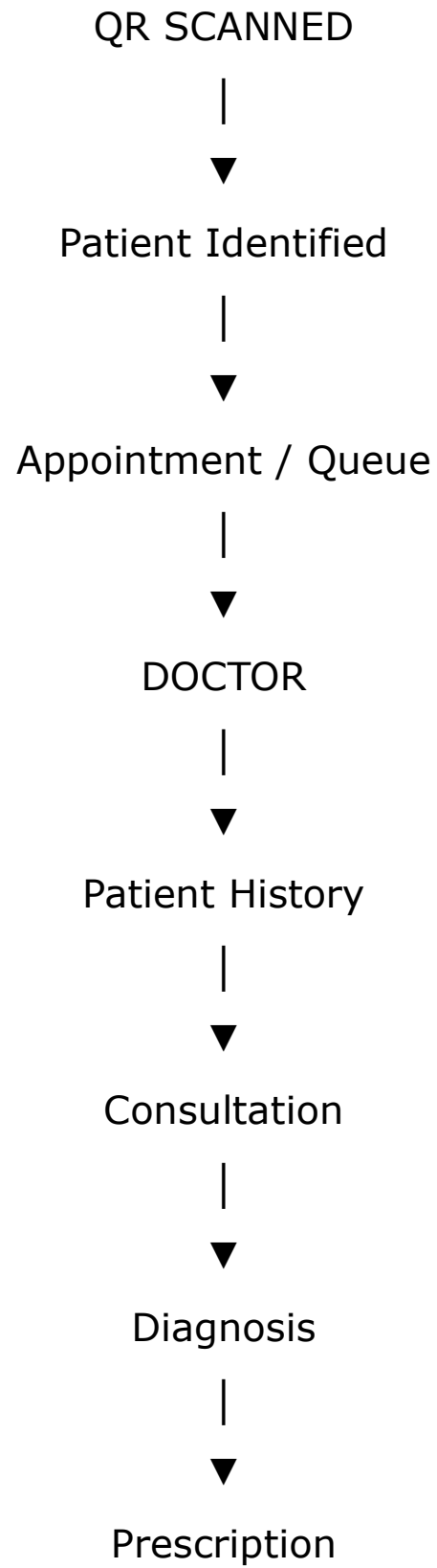
Appointments

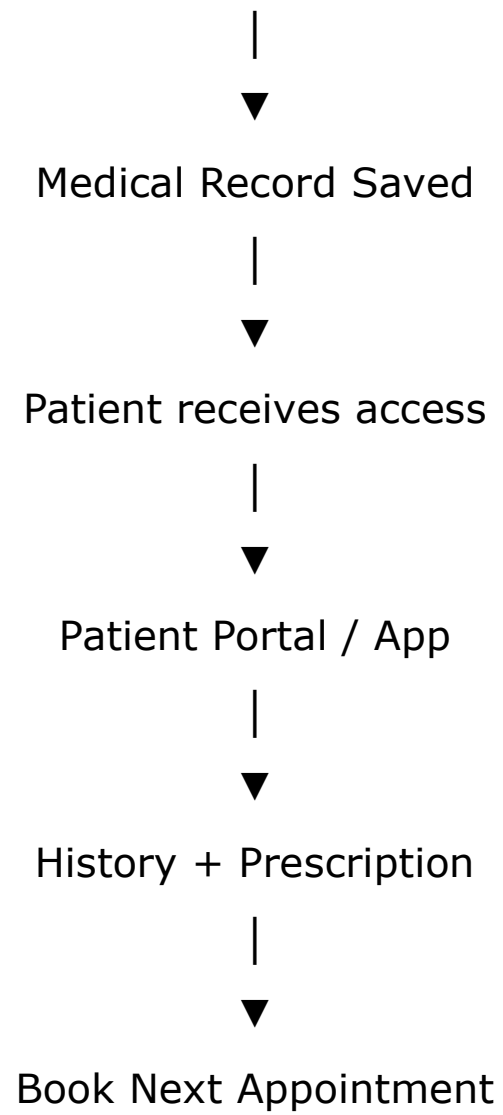
This **Role-Based Access Control (RBAC)** is essential.

13.Complete Patient Journey

- The entire system can work like this:







14.Database Structure

USERS

|

|— user_id

|— name

|— phone

|— email

|— password/OTP

|— role

|— status

PATIENTS

|

|— patient_id

|— user_id

|— name

|— DOB

|— gender

|— blood_group

|— height

└─ weight
└─ address
└─ emergency_contact
└─ allergies
└─ created_at

DOCTORS

|
└─ doctor_id
└─ name
└─ specialization
└─ qualification
└─ department
└─ availability
└─ consultation_fee

APPOINTMENTS

|
└─ appointment_id
└─ patient_id

- |— doctor_id
- |— date
- |— time
- |— status
- └— token_number

VISITS

- |
- |— visit_id
- |— patient_id
- |— doctor_id
- |— appointment_id
- |— visit_date
- |— symptoms
- |— diagnosis
- |— notes
- └— vitals

PRESCRIPTIONS

|

└─ prescription_id

└─ visit_id

└─ patient_id

└─ doctor_id

└─ diagnosis

└─ instructions

└─ created_at

PRESCRIPTION_ITEMS

|

└─ prescription_id

└─ medicine_name

└─ dosage

└─ frequency

└─ duration

└─ instructions

MEDICAL_DOCUMENTS

|

|— document_id

|— patient_id

|— visit_id

|— document_type

|— file_url

|— created_at

QR_CARDS

|

|— card_id

|— patient_id

|— token

|— issued_date

|— status